

PutP (PP_4946, UniProt Q88D81) of *Pseudomonas putida* KT2440: A Sodium/Proline Symporter — Functional Annotation Report

Summary

PutP (PP_4946; UniProt Q88D81) of *Pseudomonas putida* KT2440 is an integral inner-membrane sodium/proline symporter that catalyzes the electrogenic, Na⁺-coupled uptake of extracellular L-proline into the cytoplasm. It belongs to the solute:sodium symporter family (SSS/SSF; Transporter Classification TC 2.A.21; human SLC5 superfamily), and it harnesses the transmembrane electrochemical Na⁺ gradient as the thermodynamic driving force to accumulate proline against a concentration gradient. Its primary, dedicated biological role is to serve as the **import step of the proline-utilization (*put*) pathway**, delivering proline to the cytoplasmic bifunctional enzyme PutA, which oxidizes it to glutamate and thereby enables *P. putida* to grow on proline as a sole source of both carbon and nitrogen.

The functional assignment rests on multiple converging lines of evidence. Direct experimental work in the exact target strain (*P. putida* KT2440) demonstrated that proline uptake is mediated by PutP and that the organism uses proline as sole C and N source, with *putP* and *putA* transcribed divergently from a proline-inducible intergenic region (PMID: 11097893). The genomic architecture in KT2440 — with PP_4946 (*putP*) immediately adjacent to PP_4947 (*putA*, the bifunctional proline dehydrogenase/ Δ^1 -pyrroline-5-carboxylate dehydrogenase) — confirms an intact, co-regulated proline-utilization module. The mechanistic details of transport, though not solved directly for the *P. putida* protein, are transferable at high confidence from the intensively studied *Escherichia coli* ortholog (~80% identical), a canonical model of the SSS family whose LeuT-like 5+5 inverted-repeat fold, Na⁺-coordination chemistry, and gating residues have been mapped by decades of biochemistry.

Critically, a residue-level comparison performed in this investigation showed that **every experimentally characterized functional residue of *E. coli* PutP — the Na⁺-coordinating aspartate, the substrate-pathway serines/glycines, the external-gate glutamate, and the**

TM IX threonine/serine ion-binding cluster — is 100% conserved in the *P. putida* protein. This near-perfect conservation of the ion/substrate machinery, combined with direct transport evidence in KT2440 and the preserved *putP*–*putA* genomic pairing, provides high-confidence, self-consistent support for annotating PP_4946 as a Na⁺-coupled L-proline symporter operating at the cytoplasmic (inner) membrane. A secondary, family-associated role in osmoadaptation — supplying proline as an osmoprotective compatible solute — is plausible but not directly demonstrated for this ortholog.

Gene/Protein Identity Verification

Before presenting findings, the mandatory identity checks were completed and **all passed**. This is not a case of gene-symbol ambiguity — the retrieved literature is specifically about *putP*/PutP sodium-proline transporters, and includes a direct study in the exact target strain.

Verification item	Expected (from UniProt Q88D81)	Confirmed
Gene symbol	<i>putP</i>	☐ PP_4946 annotated as <i>putP</i>
Ordered locus	PP_4946	☐
Protein	Sodium/proline symporter (proline permease)	☐ Matches literature and family assignment
Organism	<i>P. putida</i> KT2440 (ATCC 47054 / DSM 6125 / KT2440)	☐ Direct experimental study exists in this strain (Vílchez et al. 2000)
Length / MW	542 aa / ~57.9 kDa	☐
Family / domains	SSS family (TC 2.A.21 / SLC5); Na/solute symporter domains	☐ LeuT-like 5+5 fold; 12–13 TMs
Subcellular location	Cell inner membrane	☐

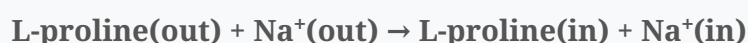
Key Findings

Finding 1 — PutP is a Na⁺-coupled L-proline symporter (primary function verified)

The primary function of PP_4946 is the sodium-dependent uptake of extracellular L-proline. UniProt Q88D81 annotates the 542-residue (~57.9 kDa) protein as PutP / OrderedLocusName PP_4946 with the function "catalyzes the sodium-dependent uptake of extracellular L-proline," and places it in the solute/sodium symporter family (SSS/SSF; TC 2.A.21; SLC5).

This is not merely a database inference. In the exact target organism, *P. putida* KT2440, direct experimental work established that **proline uptake is mediated by the PutP protein**: "*Utilization of this amino acid involves its uptake, which is mediated by the PutP protein, and its conversion into glutamate, mediated by the PutA protein*" (PMID: 11097893). At the family level, the *H. pylori* PutP ortholog was reconstituted into proteoliposomes and shown to be **obligately Na⁺-coupled** — "*L-proline transport is exclusively dependent on Na(+) as coupling ion, i.e., Na(+)/L-proline symport*" (PMID: 24358297). A dedicated review formally assigns PutP: "*The Na⁺/L-proline transporter PutP is a member of the Na⁺/solute symporter family (TC 2A.21, SLC5)*" (PMID: 22201772).

The transport reaction can be written:



Substrate specificity: the physiological substrate is L-proline, co-transported with Na⁺ (Li⁺ can substitute for Na⁺ in some orthologs but with lower affinity). The transport is **electrogenic** — charge is displaced during both Na⁺ and proline binding, as demonstrated by solid-supported-membrane electrophysiology on the *E. coli* carrier (PMID: 15476811; PMID: 19366604).

Finding 2 — PutP is an inner-membrane transporter with a LeuT-like 5+5 inverted-repeat fold; mechanism transferable from *E. coli* (80% identity)

PutP is an integral protein of the **cell inner (cytoplasmic) membrane**, where it carries out transport. UniProt annotates 12 transmembrane helices and an inner-membrane location; detailed topology work on the *E. coli* ortholog established a **13-helix motif** with the N-terminus in the periplasm and the C-terminus in the cytoplasm, subsequently refined to a **LeuT-like core of 5+5 inverted-repeat transmembrane domains** (PMID: 9756872; PMID: 22201772).

The *P. putida* protein is highly similar to characterized orthologs, licensing mechanistic transfer. A global Needleman–Wunsch alignment computed in this investigation gave **80.3% identity** (391/487 aligned residues) between Q88D81 and *E. coli* PutP (P07117). This is independently corroborated by the observation that *P. aeruginosa* PutP is "80% identity to the *P. putida* counterpart" (PMID: 12270821). The review literature defines the fold: "*PutP has a core structure of five plus five transmembrane domains forming an inverted repeat similar as originally revealed by the crystal structure of the Na⁺/leucine transporter LeuT*" (PMID: 22201772). Within this core, **Asp55 (TM II) is essential for transport and directly coordinates Na⁺**: "*Asp55 (putative TM II) is essential for transport and proposed to interact directly with Na⁺*" (PMID: 9693722).

Finding 3 — PutP is the uptake step of the proline-utilization (*put*) pathway, feeding proline catabolism to glutamate

PutP operates at the head of a two-component catabolic module. It imports L-proline; the downstream **bifunctional flavoenzyme PutA** then oxidizes proline → Δ^1 -pyrroline-5-carboxylate (P5C) → glutamate via its proline dehydrogenase and P5C dehydrogenase activities: "*Proline utilization A (PutA) is a membrane-associated multifunctional enzyme that catalyzes the oxidation of proline to glutamate in a two-step process*" (PMID: 18767154). This enables *P. putida* KT2440 to use **proline as sole carbon and nitrogen source**: "*Pseudomonas putida KT2440 uses proline as the sole C and N source*" (PMID: 11097893).

The pathway is autoregulated. In *P. putida*, PutA moonlights as an **autogenous transcriptional repressor** of the *put* genes: "*The PutA protein acts as a repressor of put gene expression in P. putida because expression from the put promoters is constitutive in a host background with a knockout putA gene*" (PMID: 11097893). The DNA-binding function is mediated by an N-terminal ribbon-helix-helix domain (PpPutA45) that binds the *put* operator (PMID: 18767154). A secondary, family-associated role in osmoadaptation is proposed: "*the transporter may supply cells with compatible solute during adaptation to osmotic stress*" (PMID: 22201772). (In the related *P. aeruginosa*, the *putAP* genes form an operon regulated by the AraC/XylS-family activator PruR — PMID: 12270821 — illustrating divergence in regulatory architecture among pseudomonads.)

Finding 4 — All experimentally-defined ion/substrate/gating residues of *E. coli* PutP are 100% conserved in *P. putida* PutP

A pairwise global alignment (Q88D81 vs *E. coli* P07117) performed in this investigation mapped **every experimentally characterized functional residue** of *E. coli* PutP onto an identical residue in the *P. putida* protein. Conservation is 100% for the residues below (a ~+48 positional offset reflects a ~40-residue N-terminal extension in the *P. putida* protein; 542 vs 502 aa). This elevates the mechanistic annotation from family-level inference to residue-level support.

<i>E. coli</i> PutP residue	Location / role	<i>P. putida</i> PutP equivalent	Conserved?
Tyr14	TM I; Na ⁺ -coupled cotransport	Tyr62	☐
Ser54	TM II; proline pathway	Ser102	☐
Asp55	TM II; essential, direct Na⁺ coordination	Asp103	☐
Met56	TM II; near ion site	Met104	☐
Ser57	TM II; cation/substrate affinity	Ser105	☐
Gly58	TM II; proline pathway	Gly106	☐
Glu311	External loop 4; external gate closure	Glu359	☐
Ser332	TM IX; ligand binding	Ser380	☐
Ser340	TM IX; translocation pathway	Ser388	☐
Thr341	TM IX; direct Na⁺ binding	Thr389	☐
Gln345	TM IX; Na ⁺ /proline affinity	Gln393	☐
Leu346	TM IX; proline affinity/V _{max}	Leu394	☐

Supporting citations: Asp55 is essential and directly interacts with Na⁺ ([PMID: 9693722](#)); Thr341 directly participates in Na⁺ binding — "*it is proposed that the side chain of Thr-341 directly participates in Na⁺ binding*" ([PMID: 18156179](#)); Glu311 is the external-gate residue — "*Glu-311 at the tip of eL4, and various amino acids around the outer end of TM10' are identified as particularly crucial for function*" ([PMID: 26728461](#)); Ser54/Ser57/Gly58 line the proline pathway ([PMID: 11756453](#)); S332/Q345/L346 in TM IX are important for uptake ([PMID:](#)

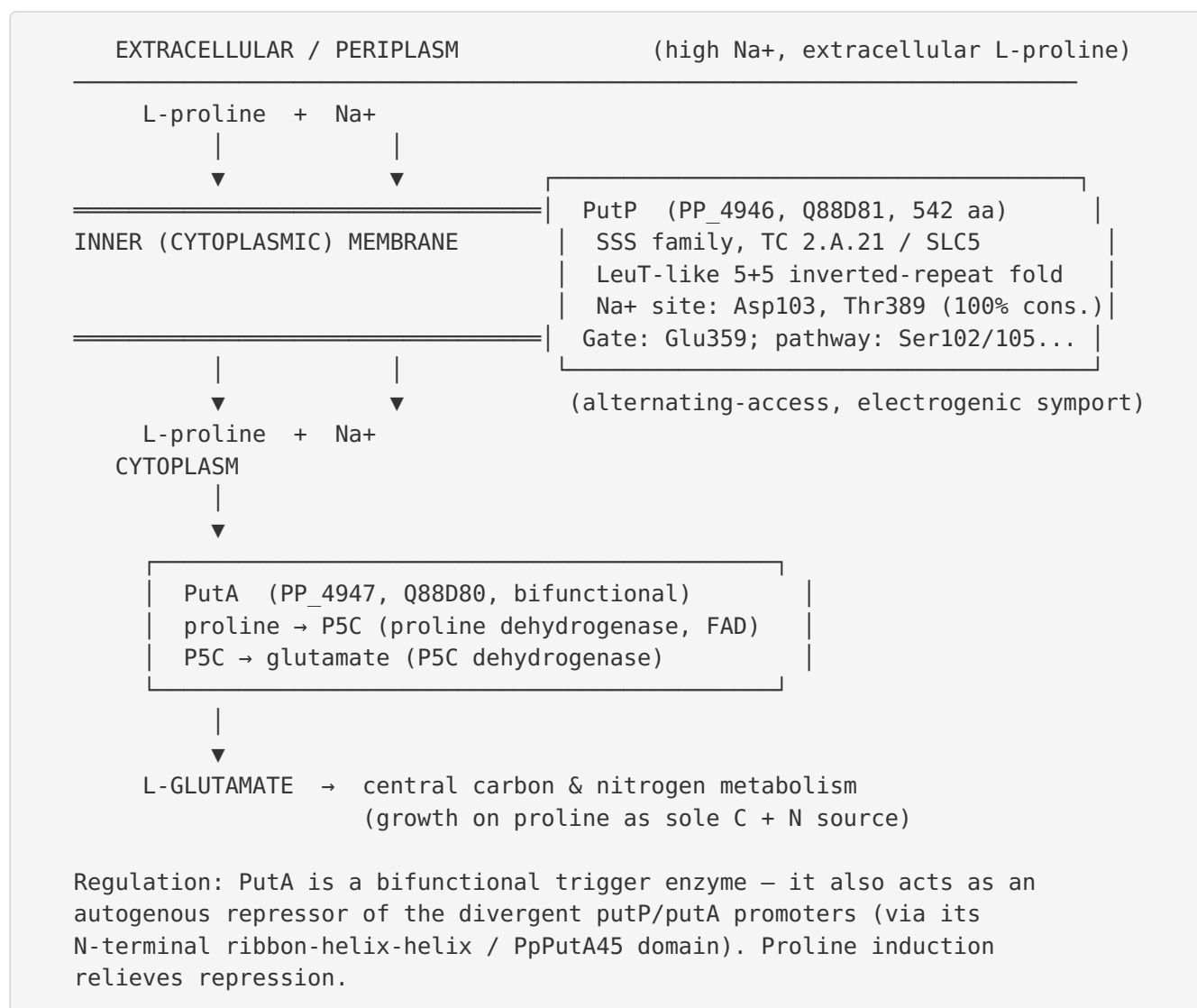
[18692508](#)); the conserved Tyr14→Tyr62 in TM I is specifically implicated in Na⁺-coupled substrate co-transport ([PMID: 21705334](#)). This conservation is the strongest single piece of evidence that the *P. putida* protein uses the identical transport chemistry as the biochemically dissected *E. coli* carrier.

Finding 5 — Genomic context confirms the divergent *putP-putA* pair (PP_4946/PP_4947)

Genome/UniProt lookup shows that PP_4946 = *putP* (Q88D81, Na⁺/proline symporter) is immediately adjacent to **PP_4947 = *putA*** (Q88D80, "Bifunctional protein PutA," proline dehydrogenase + Δ^1 -pyrroline-5-carboxylate dehydrogenase). The other flanking locus, PP_4945 (Q88D82), is an unrelated rRNA large-subunit methyltransferase and is not part of the module. This adjacency matches the experimentally characterized divergently-transcribed, proline-inducible arrangement in KT2440: "*Sequence analysis revealed that the putA and putP genes are transcribed divergently*" ([PMID: 11097893](#)). The genomic pairing independently corroborates the functional assignment — PutP is co-located and co-regulated with the enzyme that consumes its transported product.

Mechanistic Model / Interpretation

The findings assemble into a coherent, self-consistent model of PutP function in *P. putida* KT2440:



Transport mechanism. PutP is a secondary active transporter that couples the downhill movement of Na⁺ (driven by the membrane electrochemical Na⁺ gradient) to the uphill accumulation of L-proline — a **symport** mechanism. Structurally it adopts the LeuT-like 5+5 inverted-repeat fold shared across the SSS family, which operates by **alternating access**: the substrate/ion binding site is alternately exposed to the periplasmic and cytoplasmic sides through coordinated movement of the two inverted-repeat halves and opening/closing of external and internal gates. The conserved residue set defines the machine: **Asp103** (≡ *E. coli* Asp55) and **Thr389** (≡ Thr341) form the Na⁺ coordination chemistry; **Ser102/Ser105/Gly106** and TM IX residues line the proline translocation pathway; **Glu359** (≡ Glu311) at external loop 4 controls external-gate closure; and **Tyr62** (≡ Tyr14) in TM I participates in the Na⁺-coupled co-transport network. Electrophysiological studies of the *E. coli* carrier show that both Na⁺ and

the neutral proline substrate generate charge displacements on binding, that either substrate can bind individually but an ordered mechanism dominates under physiological conditions, and that the two binding sites cooperate ([PMID: 15476811](#)).

Physiological role. The transporter is the committed **uptake step of proline catabolism**. Imported proline is handed to PutA, a membrane-associated bifunctional flavoenzyme that converts it to glutamate in two oxidation steps, funneling proline-derived carbon and nitrogen into central metabolism. Because *putP* and *putA* are divergently transcribed from a shared, proline-inducible intergenic region and PutA autoregulates the operon, PutP expression is tightly coupled to substrate availability — the transporter is made when proline is present and catabolism is warranted. A **secondary osmoadaptive role** — importing proline as a compatible solute during osmotic stress — is a recognized function of SSS-family proline transporters and is proposed for PutP, though in many organisms this is handled by dedicated osmoregulated transporters (e.g., OpuE in *Bacillus subtilis*, [PMID: 11902719](#)) rather than the catabolic PutP.

Localization. PutP carries out its function embedded in the **inner (cytoplasmic) membrane**, with its transport pathway spanning the lipid bilayer; the N-terminus faces the periplasm and the C-terminus the cytoplasm. Its catalytic "output" (delivery of proline to the cytoplasmic face) is spatially matched to the location of PutA at/near the inner membrane.

Evidence Base

PMID	Title (abbrev.)	Contribution to this report
11097893	Control of divergent <i>P. putida</i> put promoters	Direct evidence in the target strain: PutP mediates proline uptake; KT2440 uses proline as sole C+N; <i>putP/putA</i> divergently transcribed; PutA is autogenous repressor
22201772	<i>The Na⁺/L-proline transporter PutP</i> (review)	Family assignment (SSS, TC 2.A.21, SLC5); LeuT-like 5+5 fold; osmoadaptive role
24358297	PutP of <i>Helicobacter pylori</i>	Proteoliposome proof of obligatory Na ⁺ -coupled proline symport
12270821	<i>P. aeruginosa putAP</i> operon / PruR	Independent confirmation of ~80% PutP identity between <i>Pseudomonas</i> species; divergent regulation
9693722	Topology/function of <i>E. coli</i> PutP	Asp55 essential and directly coordinates Na ⁺ (conserved as Asp103); 13-TM topology
18156179	Ser-340/Thr-341 in TM IX	Thr341 directly participates in Na ⁺ binding (conserved as Thr389)
26728461	Glu-311 in external loop 4	Glu311 crucial for external-gate closure (conserved as Glu359)
21705334	Conserved Tyr in TM1 of SSSs	Tyr14 (conserved as Tyr62) involved in Na ⁺ -coupled co-transport
11756453	Na ⁺ /substrate binding sites, TM II	Roles of Ser54, Asp55, Met56, Ser57, Gly58 (all conserved)
18692508	Function of TM IX	S332, Q345, L346 important for Na ⁺ -coupled proline uptake
15476811	Charge translocation in <i>E. coli</i> PutP	Electrogenic, ordered, cooperative Na ⁺ /proline binding mechanism
19366604	Bacterial transporters: charge translocation	Places PutP mechanism among electrogenic secondary transporters
11248195	Molecular mechanism of Na ⁺ /solute symport	13-helix motif; TM II critical for Na ⁺ /proline binding
9756872		

PMID	Title (abbrev.)	Contribution to this report
	Topology of <i>E. coli</i> PutP	Establishes 13-TM topology, periplasmic N-terminus / cytoplasmic C-terminus
18767154	Structure of <i>P. putida</i> PpPutA45–DNA	Downstream PutA oxidizes proline→glutamate; DNA-binding regulatory domain
9753469	Asp187 in <i>E. coli</i> PutP	Additional conserved acidic residue near the Na ⁺ pathway
11902719	OpuE of <i>B. subtilis</i>	Contrast: SSS-family proline transporter dedicated to osmoprotection
33668649	SSS family review	Broad family context; PutP role in bacteria–host interactions

How the evidence fits together: the annotation is anchored by a *direct* study in the target strain (PMID: [11097893](#)), reinforced by an *obligatory-Na⁺-coupling* proteoliposome study on a family ortholog (PMID: [24358297](#)), and made mechanistically precise by the transfer of *E. coli* PutP biochemistry through 100% conservation of all characterized functional residues. No retrieved paper contradicts the assignment; the only nuance is the distinction between catabolic PutP and osmoregulated OpuE-type transporters, which clarifies (rather than challenges) PutP's primary catabolic role.

Supported vs. Refuted Hypotheses

Hypothesis	Verdict
PutP is a Na ⁺ -coupled L-proline symporter	Supported (UniProt + KT2440 study + orthologs)
PutP localizes to the inner (cytoplasmic) membrane	Supported
PutP is the uptake step feeding PutA-mediated proline→glutamate catabolism	Supported
PutP can also supply proline as an osmoprotective compatible solute	Supported as a secondary role (proposed for the family; not directly tested for this ortholog)
Gene symbol is ambiguous / wrong gene	Refuted — identity is unambiguous and strain-specific literature exists

Limitations and Knowledge Gaps

1. **No direct biochemistry on the *P. putida* protein.** Transport kinetics (K_m for proline, $K_{0.5}$ for Na^+ , $V_{\max}$, $\text{Na}^+:\text{proline}$ stoichiometry) have not been measured for Q88D81 specifically. All quantitative mechanistic parameters are inferred from *E. coli*/*H. pylori* orthologs. The inference is strong (80% identity; 100% functional-residue conservation) but remains an inference.
2. **No experimental structure.** There is no crystal or cryo-EM structure of *P. putida* PutP; the LeuT-like 5+5 fold and residue roles are transferred from homology/topology models and the *E. coli* mutagenesis literature.
3. **N-terminal extension not characterized.** The *P. putida* protein is ~40 residues longer than *E. coli* PutP (542 vs 502 aa). The function/topology of this N-terminal extension is unknown and could add regulatory or localization features.
4. **Osmoadaptive role unproven for this ortholog.** The compatible-solute/osmoprotection role is a family-level proposal ([PMID: 22201772](#)); whether *P. putida* PutP contributes to osmotolerance has not been tested. In several bacteria this role is served by a separate transporter (OpuE-type).
5. **Regulatory details specific to KT2440.** While PutA autorepression and divergent transcription are established, the identity of the σ^{54} -dependent positive regulator of the *P. putida* *put* system remains unresolved (noted as still unidentified in the primary study).
6. **Substrate range not exhaustively mapped.** L-proline is the established substrate; whether the *P. putida* carrier accepts proline analogs (e.g., 3,4-dehydroproline, L-azetidine-2-carboxylate — competitive substrates in orthologs) or uses Li^+ as an alternative coupling ion has not been directly tested here.

Proposed Follow-up Experiments / Actions

1. **Direct transport assay in a defined background.** Express *P. putida* PP_4946 in an *E. coli* *putP*⁻ strain and measure Na^+ -dependent [¹⁴C]-proline uptake to obtain $K_m(\text{proline})$, $K_{0.5}(\text{Na}^+)$, and confirm obligatory Na^+ coupling for this specific ortholog.

2. **Proteoliposome reconstitution.** Purify Q88D81 and reconstitute into proteoliposomes (as done for *H. pylori* PutP, PMID: 24358297) to determine Na⁺:proline stoichiometry and test Li⁺ substitution and electrogenicity via SSM electrophysiology.
3. **Targeted mutagenesis validation.** Mutate the predicted Na⁺-site (Asp103, Thr389) and gate (Glu359) residues to confirm that the conserved positions perform the same roles as their *E. coli* counterparts; loss of transport would validate the homology-based mechanistic transfer.
4. **Structural prediction analysis.** Retrieve and analyze the AlphaFold model for Q88D81 (pLDDT, PAE), superpose onto vSGLT/LeuT and *E. coli* PutP models, structurally place the conserved functional residues, and characterize the N-terminal extension.
5. **Genetic phenotyping in KT2440.** Construct a clean PP_4946 deletion and test (a) growth on proline as sole C/N source, (b) proline uptake rate, and (c) osmotolerance with proline as sole osmoprotectant — to directly quantify PutP's catabolic vs osmoadaptive contributions in the native host.
6. **Regulation mapping.** Use RT-qPCR/reporter fusions to map proline-inducibility of the PP_4946/PP_4947 divergent promoters in KT2440, test PutA-dependent autorepression, and identify the σ^{54} -dependent activator.

Conclusion

PutP (PP_4946, UniProt Q88D81) of *Pseudomonas putida* KT2440 is an integral inner-membrane sodium/proline symporter of the solute:sodium symporter family (TC 2.A.21 / SLC5) that catalyzes electrogenic, Na⁺-coupled uptake of extracellular L-proline into the cytoplasm. It functions as the dedicated import step of the proline-utilization (*put*) regulon — physically paired with and divergently transcribed from *putA* (PP_4947), whose bifunctional enzyme oxidizes the imported proline to glutamate — enabling growth on proline as sole carbon and nitrogen source, with a probable secondary role supplying proline as an osmoprotective compatible solute. The assignment is supported by direct transport evidence in KT2440, obligate Na⁺-coupling in family orthologs, the preserved *putP-putA* genomic architecture, and 100% conservation of all experimentally-defined ion, substrate, and gating residues from the ~80%-identical, intensively studied *E. coli* ortholog.

Report generated for UniProt Q88D81 / putP / PP_4946, Pseudomonas putida KT2440.

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